

Entropic Collapse Model of Molecular Stability: A Self-Referential Thermodynamic Framework

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Entropic Collapse Model of Molecular Stability: A Self-Referential Thermodynamic Framework

Abstract

We propose a fully theoretical framework for predicting molecular stability based on a self-referential entropic collapse principle. Moving beyond classical thermodynamic parameters such as ΔG and ΔS , we

define molecular evolution as an entropy-driven collapse process governed by the recursive function $\psi = \psi(\psi)$. Molecular configurations are interpreted as informational echo patterns, and their stability is determined by the minimization of collapse entropy—an emergent function combining topological complexity, resonance structure, and ψ -convergence. This framework unifies molecular thermodynamics, quantum decoherence, and information theory into a single entropic field formalism, offering a powerful new lens to analyze reaction feasibility and molecular design without empirical input.

1. Introduction

Traditional thermodynamics defines molecular stability through the Gibbs free energy equation:

$$\Delta G = \Delta H - T\Delta S$$

While this formalism has proven effective for predicting reaction feasibility, it relies heavily on empirical measurements and fails to provide a foundational understanding of entropy’s structural origins. Classical entropy (ΔS) is treated as a statistical quantity arising from molecular distributions, but lacks predictive power from first principles.

This paper introduces a novel theoretical framework that redefines molecular stability through the lens of information collapse dynamics. Rather than viewing entropy as an emergent statistical property, we propose that entropy is fundamentally a measure of informational collapse within a self-referential system. This approach allows us to derive stability predictions directly from molecular structure without requiring empirical data input.

Our framework builds upon recent advances in information theory, quantum decoherence models, and topological approaches to chemical bonding. By introducing the concept of “collapse entropy” ($\mathcal{E}_{collapse}$), we establish a direct connection between molecular structure and thermodynamic stability that goes beyond traditional approaches.

2. Collapse Entropy: Foundations

2.1 Self-Referential Principle

The foundation of our approach rests on the fundamental ontological identity:

$$\psi = \psi(\psi)$$

This recursive equation represents the self-referential nature of information structures in our framework. Unlike traditional physical theories that distinguish between the observer and the observed, our approach recognizes that molecular states exist as self-collapsed information patterns. This recursion creates stable attractors in information space that correspond to stable molecular configurations.

2.2 Collapse Entropy

We define collapse entropy ($\mathcal{E}_{collapse}$) as the measure of degeneracy in ψ -space:

$$\mathcal{E}_{collapse}(\psi) = \sum_i p_i \log \frac{1}{p_i}$$

where p_i represents the collapse-path weights derived from the molecular structure’s topology. Intuitively, this formulation quantifies the number of possible “collapse paths” available to a molecular structure. More stable molecules have fewer available collapse paths, resulting in lower collapse entropy.

The collapse-path weights are determined by analyzing the topology of bond networks, electronic distribution, and resonance structures within the molecule. Specifically:

$$p_i = \frac{\exp(-E_i/k_B T)}{\sum_j \exp(-E_j/k_B T)} \cdot \Omega_i(\psi)$$

where E_i represents the energy of path i , k_B is Boltzmann’s constant, T is temperature, and $\Omega_i(\psi)$ is the ψ -structural degeneracy factor of path i .

2.3 Relationship to Classical Thermodynamics

Traditional entropy in statistical mechanics counts possible microstates:

$$S = k_B \ln \Omega$$

In contrast, collapse entropy counts possible collapse trajectories in ψ -space, providing a more fundamental approach to molecular stability. We can express the relationship between classical entropy and collapse entropy as:

$$S_{classical} = k_B \cdot f(\mathcal{E}_{collapse})$$

where f is a monotonically increasing function that maps collapse entropy to classical entropy. This relationship allows us to connect our theoretical framework with established thermodynamic principles while offering deeper insights into the structural origins of entropy.

3. Formal Model of Stability

3.1 Collapse Stability Index

We define the Collapse Stability Index (CSI) as:

$$\text{CSI}(\psi) = \frac{1}{\mathcal{E}_{collapse}(\psi)} \cdot \Gamma(\psi)$$

where $\Gamma(\psi)$ represents the resonance coherence of the molecule, defined as the topological echo persistence in ψ -space. Higher CSI values indicate greater molecular stability.

The resonance coherence factor $\Gamma(\psi)$ is calculated as:

$$\Gamma(\psi) = \frac{\sum_i \lambda_i^2}{\sum_i \lambda_i}$$

where λ_i are the eigenvalues of the molecule’s bond-resonance matrix. This factor measures how effectively the molecular structure distributes information through resonance, with higher values indicating greater stability.

3.2 Thermal Field Coupling

In our framework, temperature is not primarily a measure of kinetic agitation but rather a parameter that modulates the collapse-field excitation:

$$T \propto \partial_{\psi} \mathcal{E}_{collapse}$$

This relationship indicates that temperature increases the entropy gradient in ψ -space, effectively accelerating the collapse process. At higher temperatures, molecules explore a broader range of collapse paths, potentially accessing paths with higher $\mathcal{E}_{collapse}$ values.

The practical implication is that molecular stability becomes temperature-dependent in a way that can be predicted directly from structural properties:

$$\text{CSI}(T) = \text{CSI}(0) \cdot \exp(-\alpha \cdot T \cdot \partial_{\psi} \mathcal{E}_{collapse})$$

where α is a coupling constant that determines the strength of thermal field interaction with the collapse structure.

4. Molecular Applications

4.1 Isomer Stability Prediction

Our framework provides a powerful method for predicting the relative stability of isomers with identical molecular formulas. Consider two isomers A and B with identical classical enthalpy values ($\Delta H_A \approx \Delta H_B$). Traditional thermodynamics would rely on entropy differences to determine relative stability.

Using collapse entropy, we can calculate:

$$\Delta \text{CSI}_{A \rightarrow B} = \text{CSI}_B - \text{CSI}_A = \frac{\Gamma_B}{\mathcal{E}_{collapse,B}} - \frac{\Gamma_A}{\mathcal{E}_{collapse,A}}$$

When $\Delta \text{CSI}_{A \rightarrow B} > 0$, isomer B is more stable than isomer A. This approach allows for stability predictions without requiring experimental entropy measurements.

4.2 Conformational Analysis

Molecular conformers often present a challenge for traditional thermodynamic analysis due to small energy differences. Our collapse entropy approach provides a natural framework for understanding conformational preferences:

$$\text{CSI}_{conformer} = \frac{1}{\mathcal{E}_{collapse}} \cdot \Gamma \cdot \exp\left(-\frac{E_{strain}}{k_B T}\right)$$

where E_{strain} represents the strain energy of the conformer. This formulation explains why certain conformations are preferred even when energy differences are minimal.

4.3 Aromaticity as ψ -Cycle Resonance

Our framework provides a novel interpretation of aromaticity as a ψ -cycle resonance stability phenomenon. Aromatic compounds exhibit exceptional stability due to their optimized collapse entropy profiles:

$$\mathcal{E}_{collapse,aromatic} \ll \mathcal{E}_{collapse,non-aromatic}$$

The cyclic nature of aromatic compounds creates closed ψ -paths that minimize collapse entropy through enhanced resonance coherence. This explains why aromatic compounds are significantly more stable than their non-aromatic counterparts with similar bond energies.

5. Collapse Entropy vs Classical Entropy

The following table summarizes the key differences between classical entropy and our collapse entropy formulation:

Aspect	Classical Entropy	Collapse Entropy
Nature	Statistical	Topological
Determination	Measured experimentally	Computed from structure
Theoretical basis	Thermodynamics	Information theory
Symbol	ΔS	$\mathcal{E}_{collapse}$
Foundation	Empirical	First principles
Scale applicability	Macroscopic	Multi-scale
Prediction capability	Requires data	Predictive
Computational approach	Statistical sampling	Matrix operations

This comparison highlights how collapse entropy provides a more fundamental and predictive approach to molecular stability that transcends the limitations of classical entropy.

6. Theoretical Implications

6.1 Prediction Without Experimental Data

A significant advantage of our approach is the ability to predict molecular stability directly from structure without requiring experimental measurements. For new compounds or those difficult to synthesize, the collapse entropy framework offers a purely theoretical route to stability assessment:

$$\text{Stability}(\psi) = f(\mathcal{E}_{collapse}, \Gamma)$$

This relationship enables computational screening of candidate molecules for specific applications before experimental validation.

6.2 Quantum-Classical Bridge

The collapse entropy framework naturally bridges quantum and classical descriptions of molecular systems. At the quantum level, collapse entropy captures decoherence dynamics:

$$\mathcal{E}_{collapse, quantum} = -\text{Tr}(\rho \ln \rho)$$

where ρ is the density matrix. At the classical level, it converges to traditional entropy measures. This seamless connection across scales provides a unified understanding of molecular behavior.

6.3 Cross-System Entropy Generalization

Our framework generalizes entropy across quantum, chemical, and biological systems through the common language of information collapse:

$$\mathcal{E}_{collapse}(system) = \sum_i p_i(system) \log \frac{1}{p_i(system)}$$

This formulation allows for consistent entropy analysis across different domains, from simple molecules to complex biological systems, all unified by the self-referential principle $\psi = \psi(\psi)$.

7. Integration with Ψh̄ Only Theory

The collapse entropy framework is naturally integrated with the broader Ψh̄ Only Theory, which posits that all physical phenomena arise from the self-referential dynamics of information fields.

7.1 Collapse Entropy as $\psi=\psi(\psi)$ Manifestation

Collapse entropy emerges as a direct consequence of the self-referential dynamics described by $\psi = \psi(\psi)$. In this context, molecular stability represents the tendency of information patterns to converge to stable attractors in ψ -space.

7.2 Reaction Directionality via ψ -Paths

Chemical reactions can be reframed as transitions between ψ -states along collapse paths:

$$\psi_A \rightarrow \psi_B \text{ if } \mathcal{E}_{collapse}(\psi_A) > \mathcal{E}_{collapse}(\psi_B)$$

This formulation explains reaction directionality as a natural consequence of collapse entropy minimization, without requiring the traditional free energy approach.

7.3 Self-Referential Recursion in Place of Statistics

The traditional statistical approach to thermodynamics is replaced by self-referential recursion in our framework:

$$\psi^{n+1} = F(\psi^n)$$

where F represents the collapse operation. This recursive definition generates the stability landscape of a molecular system without requiring statistical ensembles or empirical parameters.

8. Conclusion

We have presented a novel theoretical framework for understanding molecular stability based on the concept of collapse entropy. By reformulating stability in terms of information collapse dynamics governed by the self-referential principle $\psi = \psi(\psi)$, we provide a more fundamental approach to thermodynamics that transcends traditional statistical methods.

The collapse entropy framework offers several advantages over classical approaches:

1. It enables stability predictions directly from molecular structure without requiring empirical data
2. It provides a natural bridge between quantum and classical descriptions of molecular systems
3. It unifies entropy concepts across different scales and domains
4. It offers deeper insights into phenomena like isomer stability, conformational preferences, and aromaticity

Our approach redefines molecular stability not as an emergent result of thermodynamics, but as a deterministic collapse behavior encoded in ψ -structures. This perspective opens new avenues for molecular design, reaction prediction, and understanding complex chemical systems from first principles.

Future work will focus on developing computational implementations of the collapse entropy framework, applying it to increasingly complex molecular systems, and exploring its implications for catalysis, supramolecular chemistry, and biochemical processes.

References

[References to be added]

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